Python for Supply Chain

Session 6 · Thu 3 Dec 2026

Make the numbers talk

Today's question

How do we show the finding so a manager gets it in five seconds?

A merge adds columns from another table

IN DAILY LIFE

Matching a packing list to the product catalogue.

IN EXCEL

VLOOKUP or XLOOKUP.

THE LIMIT

Count rows before and after.

```
names = sales_clean[["stock_code", "description"]]
names = names.drop_duplicates(subset="stock_code")
product_value = product_value.merge(
    names, on="stock_code", how="left")
print(len(product_value))
```

A repeated name grows the table

Step	Rows
Product table	3,066
Naive merge	3,107
After dedupe	3,066

40 product codes have two names. That adds 41 rows.

A chart is a few lines of code

IN DAILY LIFE

Drawing a chart by hand.

IN EXCEL

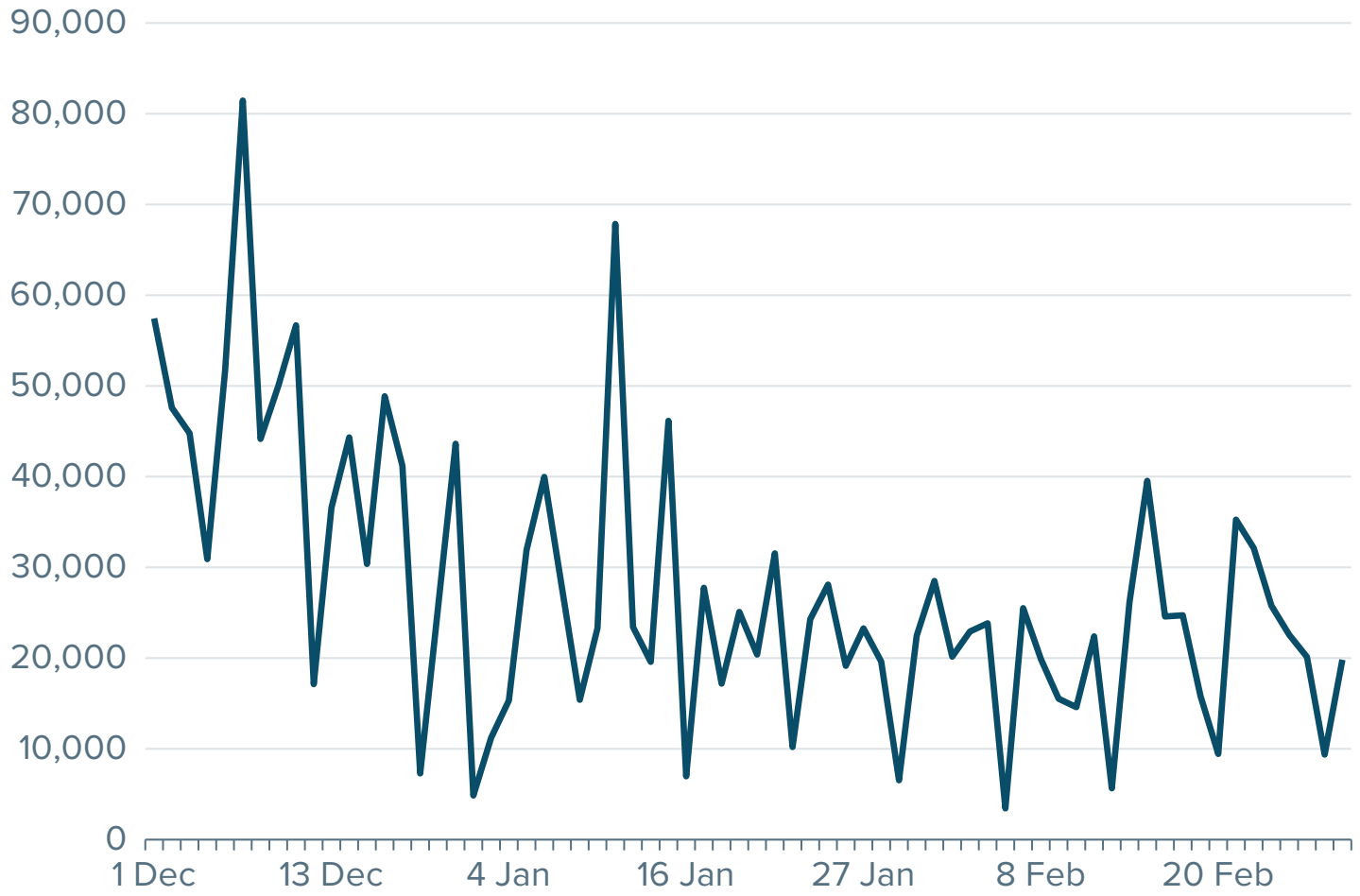
Insert Chart, then edit each part.

THE LIMIT

Every chart needs a title and axis labels.

```
plt.barh(top10["description"],
         top10["revenue"])
plt.title("Top 10: 9.6% of revenue")
plt.xlabel("Revenue (pounds)")
plt.show()
```

Revenue peaked in the first week of December



81,454.99

best day, 7 December

68

trading days

Clean slice, Dec 2010 to Feb 2011

Seaborn styles the same chart

IN DAILY LIFE

A chart template on top of matplotlib.

IN EXCEL

Chart styles.

THE LIMIT

The same titles and labels still work.

```
import seaborn as sns
top = country_revenue.head(5).reset_index()
sns.barplot(data=top, x="revenue", y="country")
plt.title("Outside the UK, the Netherlands lead")
plt.show()
```

Now in Colab

Notebook	s6_numbers_talk.ipynb
In class	Merge, then two charts
At the end	Case hunt explained

Three things you can do now

Join tables

merge and check rows

Draw a chart

title, labels, show

Make it readable

title says the finding

Homework and daily recall

Homework

1. Finish the top 10 chart
2. Try the seaborn pattern
3. Start the case hunt
4. Install Power BI Desktop

Daily recall

1. Three questions Friday
2. Three questions Saturday
3. Three questions Sunday